

IDS 702 HW 5

KEY

Instructions: Use this template to complete your assignment. When you click “Render,” you should get a PDF document that contains both your answers and code. You must show your work/justify your answers to receive credit. Submit your rendered PDF file on Gradescope. **Remember to render frequently**, as this will help you to catch errors in your code before the last minute. You must show your work for all problems, and you must provide a written answer for all problems; it is insufficient to just show the code.

Add your name in the Author section in the header

Load data

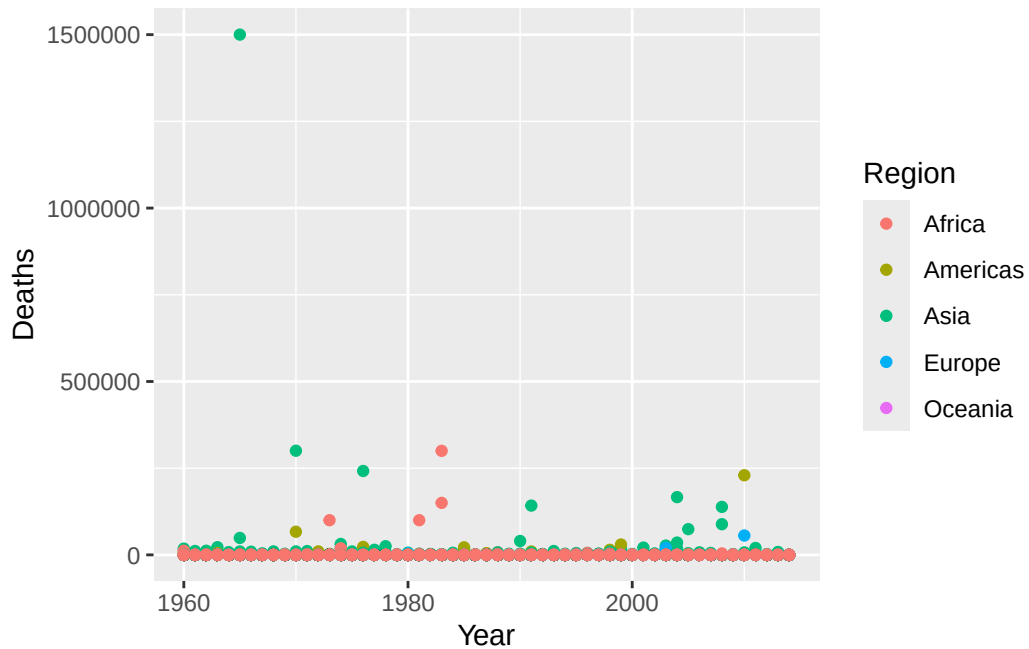
```
library(MASS)
library(tidyverse)
library(glmTMB)
## NOTE: Run install.packages("glmTMB", type = "source") IN CONSOLE

disaster_deaths <- read.csv("https://raw.githubusercontent.com/anlane611/datasets/refs/heads,
```

1

a.

```
ggplot(disaster_deaths, aes(x=Year, y=Deaths, col=Region))+
  geom_point()
```



We see an outlier that stands out from the rest of the data, and we see that many of the points are very small or zero. It is difficult to see a clear trend over time.

b.

```
disaster_deaths |>
  mutate(event = ifelse(Deaths>0,1,0)) |>
  count(event) |>
  mutate(prop=n/sum(n))
```

	event	n	prop
1	0	8109	0.712253
2	1	3276	0.287747

8109 (71%) of the rows are 0, and 3276 (29%) of the rows are nonzero.

2

a.

```
disaster_pois_mod <- glm(Deaths ~ Region*Year,
                        data=disaster_deaths,
                        family="poisson")

summary(disaster_pois_mod)
```

Call:

```
glm(formula = Deaths ~ Region * Year, family = "poisson", data = disaster_deaths)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.183e+01	1.358e-01	234.39	<2e-16 ***
RegionAmericas	-8.656e+01	2.264e-01	-382.29	<2e-16 ***
RegionAsia	4.798e+01	1.536e-01	312.33	<2e-16 ***
RegionEurope	-1.862e+02	4.736e-01	-393.12	<2e-16 ***
RegionOceania	-7.373e+01	1.496e+00	-49.27	<2e-16 ***
Year	-1.317e-02	6.846e-05	-192.42	<2e-16 ***
RegionAmericas:Year	4.337e-02	1.138e-04	381.24	<2e-16 ***
RegionAsia:Year	-2.343e-02	7.748e-05	-302.33	<2e-16 ***
RegionEurope:Year	9.270e-02	2.367e-04	391.67	<2e-16 ***
RegionOceania:Year	3.515e-02	7.510e-04	46.81	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 56988200 on 11384 degrees of freedom
Residual deviance: 49614623 on 11375 degrees of freedom
AIC: 49633234

Number of Fisher Scoring iterations: 12

```
#effect for Africa per year
exp(coef(disaster_pois_mod)[6])
```

Year
0.9869141

```
#effect for other regions
exp(coef(disaster_pois_mod)[6]+coef(disaster_pois_mod)[7:10])
```

RegionAmericas:Year	RegionAsia:Year	RegionEurope:Year	RegionOceania:Year
1.0306537	0.9640637	1.0827807	1.0222237

The interaction coefficients in this case assess differences in trends over time for different regions. The exponentiated coefficient estimate for Year is 0.99, which means that the expected count of deaths decreases by 1% each year **in Africa**. Asia similarly experiences a 4% decrease. In contrast, the Americas experience a 3% increase per year ($\exp(-0.013+.04)=1.03$), Oceania experiences a 2% increase, and Europe experiences the largest change in magnitude at 8% per year. This seems substantial for a per-year basis, but we should assess assumptions to see if this model is suitable.

b.

```
disaster_nb_mod <- MASS::glm.nb(Deaths ~ Year*Region,
                                data=disaster_deaths)

AIC(disaster_pois_mod, disaster_nb_mod)
```

	df	AIC
disaster_pois_mod	10	49633234.07
disaster_nb_mod	11	53508.88

The AIC is much lower for the negative binomial model, which gives evidence of overdispersion in the model. (We can also see that the theta value in the NB model is very small, but this is not required to answer the question).

3

```
disaster_pois_mod_rate <- glm(Deaths ~ Region*Year,
                              offset=log(Population),
                              data=disaster_deaths,
                              family="poisson")

summary(disaster_pois_mod_rate)
```

Call:

```
glm(formula = Deaths ~ Region * Year, family = "poisson", data = disaster_deaths,  
     offset = log(Population))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	6.891e+01	1.362e-01	505.98	<2e-16 ***
RegionAmericas	-1.104e+02	2.283e-01	-483.39	<2e-16 ***
RegionAsia	3.107e+01	1.545e-01	201.19	<2e-16 ***
RegionEurope	-2.359e+02	4.766e-01	-495.02	<2e-16 ***
RegionOceania	-9.245e+01	1.501e+00	-61.59	<2e-16 ***
Year	-3.996e-02	6.866e-05	-581.94	<2e-16 ***
RegionAmericas:Year	5.516e-02	1.147e-04	480.95	<2e-16 ***
RegionAsia:Year	-1.582e-02	7.791e-05	-203.12	<2e-16 ***
RegionEurope:Year	1.174e-01	2.382e-04	493.05	<2e-16 ***
RegionOceania:Year	4.567e-02	7.535e-04	60.61	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 40098782 on 10749 degrees of freedom
Residual deviance: 36105031 on 10740 degrees of freedom
(635 observations deleted due to missingness)
AIC: 36123540

Number of Fisher Scoring iterations: 11

```
#risk for Africa per year  
exp(coef(disaster_pois_mod_rate)[6])
```

Year
0.9608327

```
#risk for other regions  
exp(coef(disaster_pois_mod_rate)[6]+coef(disaster_pois_mod_rate)[7:10])
```

RegionAmericas:Year	RegionAsia:Year	RegionEurope:Year	RegionOceania:Year
1.0153256	0.9457482	1.0805558	1.0057293

The risk of death in a natural disaster decreases 4% each year in Africa and 5% per year in Asia. In the Americas and Europe, the risk increases by 2% and 8%, respectively, and .6% in Oceania. These numbers are very similar to what we saw with the expected counts.

4

```
disaster_deaths <- disaster_deaths |>
  mutate(Year_1 = Year - min(Year)+1)

zip_model_pois <- glmmTMB(
  Deaths ~ Region*Year_1,
  ziformula = ~ Region*Year_1,
  family = poisson,
  data = disaster_deaths
)

summary(zip_model_pois)
```

```
Family: poisson ( log )
Formula:      Deaths ~ Region * Year_1
Zero inflation: ~Region * Year_1
Data: disaster_deaths

      AIC      BIC    logLik -2*log(L)  df.resid
36235208 36235355 -18117584  36235168     11365
```

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	9.227e+00	2.108e-03	4377	<2e-16 ***
RegionAmericas	-3.115e+00	4.206e-03	-741	<2e-16 ***
RegionAsia	5.089e-01	2.287e-03	222	<2e-16 ***
RegionEurope	-5.325e+00	1.182e-02	-450	<2e-16 ***
RegionOceania	-5.452e+00	2.968e-02	-184	<2e-16 ***
Year_1	-7.160e-02	7.344e-05	-975	<2e-16 ***
RegionAmericas:Year_1	8.420e-02	1.207e-04	698	<2e-16 ***
RegionAsia:Year_1	1.367e-02	8.223e-05	166	<2e-16 ***
RegionEurope:Year_1	1.173e-01	2.707e-04	434	<2e-16 ***
RegionOceania:Year_1	7.243e-02	8.202e-04	88	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.043365	0.120635	25.228	< 2e-16 ***
RegionAmericas	-1.449883	0.154572	-9.380	< 2e-16 ***
RegionAsia	-1.660818	0.150882	-11.007	< 2e-16 ***
RegionEurope	-0.009475	0.190579	-0.050	0.9603
RegionOceania	-0.479829	0.228987	-2.095	0.0361 *
Year_1	-0.067464	0.003249	-20.768	< 2e-16 ***
RegionAmericas:Year_1	0.039824	0.004303	9.255	< 2e-16 ***
RegionAsia:Year_1	0.037714	0.004220	8.937	< 2e-16 ***
RegionEurope:Year_1	0.007805	0.005086	1.535	0.1249
RegionOceania:Year_1	0.042580	0.006324	6.733	1.66e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

a.

```
zip_model_nb <- glmmTMB(
  Deaths ~ Region*Year_1,
  ziformula = ~ Region*Year_1,
  family = nbinom2,
  data = disaster_deaths
)

AIC(zip_model_pois, zip_model_nb)
```

	df	AIC
zip_model_pois	20	36235208.20
zip_model_nb	21	52352.35

Again, the AIC for the negative binomial model is much lower, so there is evidence of overdispersion in the poisson model.

b.

```
summary(zip_model_nb)
```

```
Family: nbinom2 ( log )
Formula: Deaths ~ Region * Year_1
```

```
Zero inflation:      ~Region * Year_1
Data: disaster_deaths
```

AIC	BIC	logLik	-2*log(L)	df.resid
52352.4	52506.5	-26155.2	52310.4	11364

Dispersion parameter for nbinom2 family (): 0.0858

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	10.011193	0.316421	31.64	< 2e-16 ***
RegionAmericas	-4.452601	0.366346	-12.15	< 2e-16 ***
RegionAsia	-1.173974	0.363977	-3.23	0.00126 **
RegionEurope	-6.571475	0.425580	-15.44	< 2e-16 ***
RegionOceania	-7.197411	0.535781	-13.43	< 2e-16 ***
Year_1	-0.118786	0.007762	-15.30	< 2e-16 ***
RegionAmericas:Year_1	0.127808	0.009273	13.78	< 2e-16 ***
RegionAsia:Year_1	0.074572	0.009180	8.12	4.55e-16 ***
RegionEurope:Year_1	0.157861	0.010476	15.07	< 2e-16 ***
RegionOceania:Year_1	0.117796	0.014226	8.28	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.79422	0.20298	18.693	< 2e-16 ***
RegionAmericas	-2.88198	0.25512	-11.296	< 2e-16 ***
RegionAsia	-2.71921	0.24605	-11.051	< 2e-16 ***
RegionEurope	-1.06784	0.31185	-3.424	0.000617 ***
RegionOceania	-2.22720	0.34471	-6.461	1.04e-10 ***
Year_1	-0.14204	0.00805	-17.644	< 2e-16 ***
RegionAmericas:Year_1	0.08988	0.01006	8.939	< 2e-16 ***
RegionAsia:Year_1	0.07604	0.01009	7.536	4.83e-14 ***
RegionEurope:Year_1	0.04267	0.01182	3.611	0.000304 ***
RegionOceania:Year_1	0.10341	0.01209	8.551	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
#exponentiate coefficient for Year (Africa)
#for conditional (negative binomial) model
exp(fixef(zip_model_nb)$cond[6])
```

```
Year_1
0.8879975
```

```
#exponentiate coefficients for other regions - conditional
exp(fixef(zip_model_nb)$cond[6]+fixef(zip_model_nb)$cond[7:10])
```

```
RegionAmericas:Year_1    RegionAsia:Year_1    RegionEurope:Year_1
      1.0090629           0.9567489           1.0398484
RegionOceania:Year_1
      0.9990101
```

```
#exponentiate coefficient for Year (Africa)
#for logistic model
exp(fixef(zip_model_nb)$zi[6])
```

```
Year_1
0.8675886
```

```
#exponentiate coefficients for other regions
exp(fixef(zip_model_nb)$zi[6]+fixef(zip_model_nb)$zi[7:10])
```

```
RegionAmericas:Year_1    RegionAsia:Year_1    RegionEurope:Year_1
      0.9491810           0.9361363           0.9054092
RegionOceania:Year_1
      0.9621117
```

Compared to the poisson model fit in number 2, the coefficient estimates in the zero-inflated conditional model are greater in magnitude, which makes sense because the zeros aren't included in that model. When there is at least one disaster death, the expected count of deaths in Africa decreases by 12% each year. The Americas and Oceania stay relatively constant, deaths in Africa decrease by 4% each year, and deaths in Europe increase 4% each year. Looking at the logistic model, the odds of **not** experiencing a disaster death in Africa decrease by about 13% each year, so it's interesting that the odds of a zero decrease but the expected count of deaths also decreases. The odds in all other regions are also less than one, indicating that the odds of experiencing a disaster death increase (because this is modeling the probability/odds of a zero).

Bonus

The independence assumption may be violated here because years that are close to each other in the same region may not be independent (this is called temporal correlation). To account for this, we can use robust standard errors like we saw in the bonus question of HW 3 part 2.